

- Explain the different types of electronic transition of methanol and methyl chloride.
- Write the postulates of VSEPR Theory? Find the steric number, molecular geometry and shape of ClF_3 , SF_4 , I_3^- and BF_3 .
- Draw the potential energy plot for orbital overlapping, and discuss the allowed and forbidden interactions according to Valence Bond Theory.
- Draw the splitting pattern of d-orbitals in octahedral and tetrahedral geometry of the complexes.
- Write the working principle of FTIR? Mention and explain the difference of stretching frequency of -OH in the free and H-bonded form.
- What is the role of reference and attenuator in UV-Vis Spectrophotometer? Explain the types of electronic transition possible in C_2H_4 and HCHO.
- Give the shape and geometry of XeF_2 .
- With the help of VBT explain hybridization, geometry and magnetic property of $[\text{NiCl}_4]^{2-}$.
- Find the steric no. and molecular geometry of BrF_3 and SF_4 . Draw the overall geometry of the compound predicted by the steric number of system.
- Define molecular spectroscopy. Explain the different types of transition states with the help of suitable energy level diagrams.
- What is Beer-Lambert Law? State its mathematical expression and significance. Draw the block/flow diagram of a UV-visible spectrophotometer.
- Draw the energy level diagram of different transition States and explain.
- What is the working principle of UV-Vis Spectroscopy? Illustrate and explain the double beam UV-Vis Spectrophotometer.

Unit 2: Chemical Kinetics & Surface Chemistry

- Derive the rate of reaction for enzyme-catalyzed reaction which proceeds through the following steps: **Step I: $E + S \leftrightarrow ES$** and **Step II: $ES \rightarrow P + E$** .
- In Langmuir adsorption isotherm A is the gas molecule and M is the surface site. **$A_{(g)} + M_{(\text{Surface})} \leftrightarrow AM$** . Calculate the fraction of surface sites occupied by gas molecules (θ).
- A reaction has rate constant of 5000 sec^{-1} at 25°C and activation energy of $6 \times 10^4 \text{ J/mol}$. At what temperature (in K) would rate constant be 10000 sec^{-1} ?